

Omar El-Banna

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EDUCATION

University of Minnesota

Bachelor of Science in Computer Science

Minneapolis, MN

Expected May 2028

- Coursework: DSA, Discrete Structures, Probability and Statistics, OOP, Linear Algebra
- Awards: University Honors Program, Presidential Scholarship, Dean's List (3x)
- GPA: 3.8/4.0

EXPERIENCE

Artificial Intelligence Researcher

January 2026 – Present

Algoverse

Remote

- Earned 3 ICML 2026 *workshop acceptances* (CompLearn, FoGen, ML for Audio) studying prior dominance in audio-visual LLMs, mentored by Stanford and UC Berkeley researchers.
- Built an end-to-end PyTorch and Hugging Face evaluation pipeline to benchmark VideoLLaMA2 and InternVideo2 across 6,300+ AVHBench samples, quantifying a 32.3% accuracy drop and 17.3% instruction failure rate under cross-modal conflict.
- Applied logit lens analysis across 1,281 conflict samples and 4 model configurations to localize a consistent snap layer at 25.5 ± 1 , showing AV-LLMs commit to memorized priors before processing cross-modal evidence at generation time.
- Designed Asymmetric Modality Dropout (AMD) ablations and conducted head shutoff experiments across layers 14-27, isolating 21 conflict-resolution heads in layers 15-18, structurally upstream of where learned priors override cross-modal computation.

Team Lead | *Certified Apple Advisor*

June 2025 – Present

Best Buy

Coon Rapids, MN

- Led a team of Sales Advisors and ranked Top 10 in market, as a result of \$550K+ quarterly revenue.
- Certified as an advisor on Apple ecosystem integration across iPhone, Mac, and iPad product lines.

PROJECTS

Reinforcement Learning on Cuphead Video Game

May 2026 – Present

- Built a custom Gymnasium environment with 12 discrete actions, frame skipping, and input gating to train a PPO agent on Cuphead boss fights, eliminating stray keypress bugs and enabling 4x faster training iterations without access to game internals.
- Trained a YOLOv8 Nano object detector on 45 labeled screenshots augmented to 109, achieving 0.99 detection confidence at 30ms inference on Apple Silicon for real-time enemy and projectile tracking.
- Engineered a multi-signal vision pipeline using template matching and HSV color analysis for HP and game state detection, replacing game memory access; achieved 0.764 explained variance with the agent clearing the first boss.

Flick

June 2026

- Built a real-time movie-night web app, with a live group session, where 5+ friends join, vote on 5 AI-picked movies, and see the winning movie pop up on every screen at once, through developing a full-stack platform with Python (FastAPI), TypeScript (Next.js), and OpenAI's GPT-4o-mini.
- Raised movie recommender accuracy from ~20% to near-perfect by validating every model output against a trusted film database before displaying to the user.
- Set up production-grade security and quality safeguards, through automated tests that detect stolen login tokens and a CI pipeline that runs security scans on every code change, by implementing encrypted login with token rotation, Google Sign-In, and a GitHub Actions workflow that lints, tests, and scans on every push.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript

Frameworks & Libraries: PyTorch, Next.js, FastAPI

Developer Tools: Git/Github, Lambda (GPU Cloud Compute), Docker, VS Code, IntelliJ

INVOLVEMENT

Minorities in STEM | *Founding Board Member*

October 2025 – Present

- * Coordinated University Events of 200+ students and worked on Front-End design for social media pages.
- * Led a team of Social Media and Event Coordinators in University of Minnesota Student Organization.